

Drive Alert : A Camera-Based Driver Drowsiness Detection and Multi-Modal Alert System

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Course:Interactive System Engineering 2



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Problem & Motivation

Driver fatigue is one of the leading causes of road accidents — yet a tired driver rarely realises how impaired they are. DriveAlert detects drowsiness early and warns the driver in time, without ever taking over the vehicle.



50 ms

monitoring cycle (30 FPS IR camera)



15% / 30%

PERCLOS thresholds → mild / severe



10 s

no-response window before emergency

The challenge

- Fatigue impairs reaction time like alcohol
- Self-assessment is unreliable
- Single-sensor systems give false alarms
- Real cabins: low light, glasses, vibration

Our goal (R0)

- Detect drowsiness from face + vehicle cues
- Warn the driver early and progressively
- Escalate to emergency only if unresponsive
- Never actuate steering, brakes or engine

Why MBSE / SysML

- One consistent model, not scattered docs
- Requirements traced to design & tests
- Structure, behavior & math in one place
- Covers all four SysML pillars (this deck)



System Concept — How DriveAlert Works



A continuous sense → analyse → decide → alert loop. The Driver Monitoring ECU fuses three fatigue cues into one drowsiness score, then drives a tiered, multi-modal response.



1 · Sense

IR camera @ 30 FPS captures the driver's face; steering & lane data add secondary confirmation.



2 · Analyse

Compute PERCLOS eye-closure, detect yawns & head-nods in parallel, then fuse.



3 · Decide

score = $0.6 \cdot \text{PERCLOS} + 0.25 \cdot \text{yawn} + 0.15 \cdot \text{nod}$ → classify Awake / Mild / Severe.



4 · Alert

Tiered visual → audio → haptic; escalate to eCall if no response in 10 s.



Guiding principle — the driver stays in control

DriveAlert is an advisory system. It warns, advises and (in emergencies) calls for help, but it never steers, brakes or accelerates. The human always retains full authority (R4).



Modelling Approach — Four SysML Pillars

The whole system is captured in one SysML model organised into four packages. Eight diagrams (shown in this deck) together cover requirements, structure, behavior and parametrics.



1 · Requirements

Requirements Diagram (R0–R6)



2 · Structure

Block Definition (BDD) · Internal Block (IBD)



3 · Behavior

Use Case · Activity · Sequence · State Machine



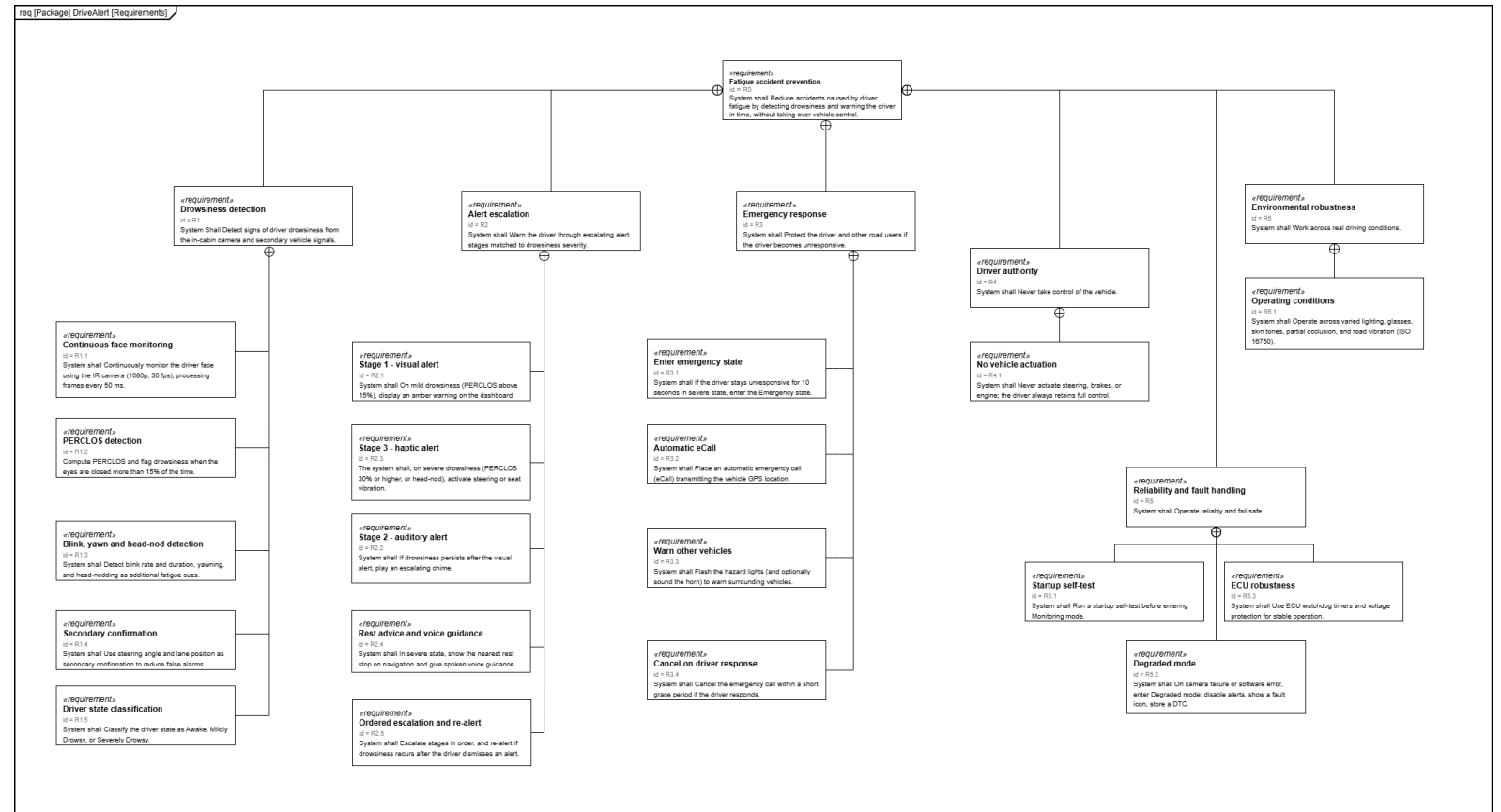
4 · Parametrics

Parametric Diagram (PERCLOS, fusion, thresholds)

Requirements Model

From mission to detail

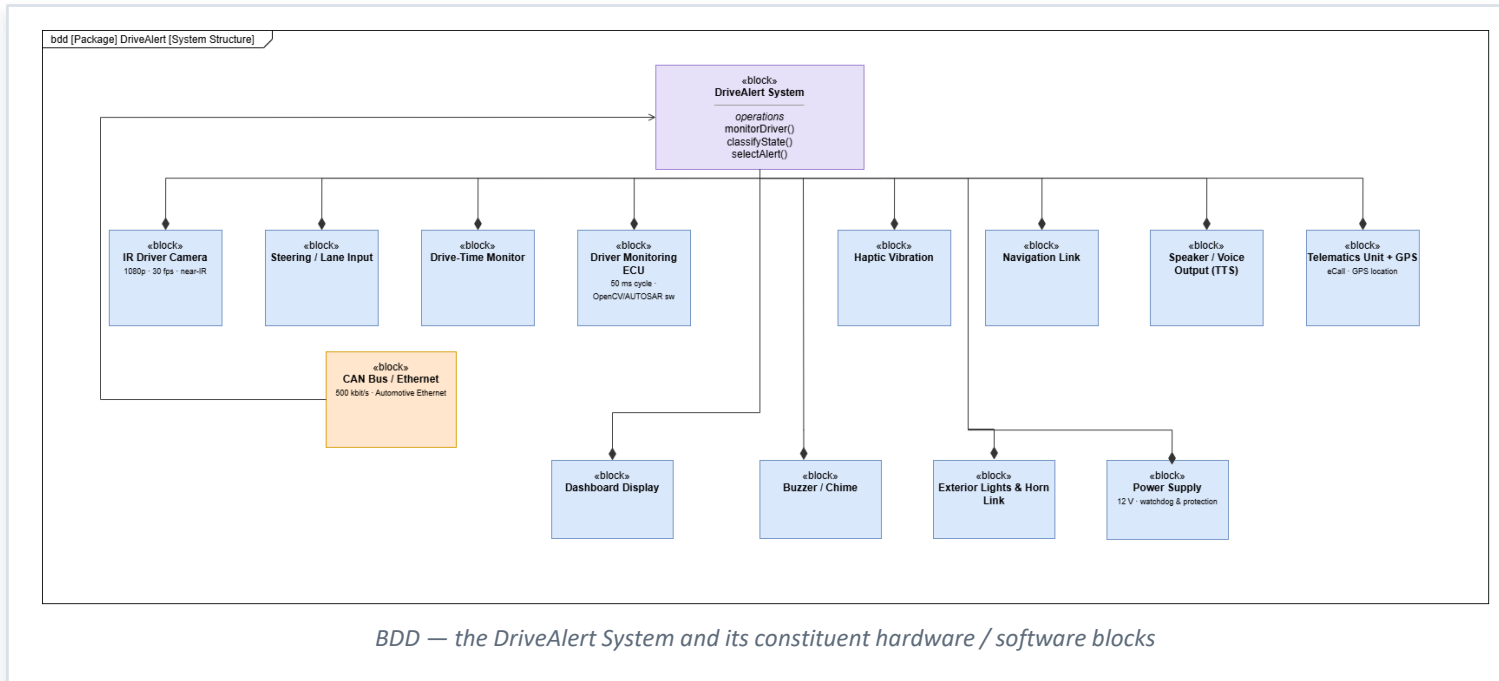
- R0 Fatigue accident prevention (top goal)
- R1 Drowsiness detection — PERCLOS, blink, yawn, head-nod, state classification
- R2 Alert escalation — visual → audio → haptic, ordered & re-alert
- R3 Emergency response — eCall + GPS, warn others, cancel on response
- R4 Driver authority — no vehicle actuation
- R5 Reliability — self-test, degraded mode, ECU watchdog
- R6 Environmental robustness — lighting, glasses, vibration (ISO 16750)



Requirements Diagram — R0 mission decomposed into seven branches (R1–R6) with «requirement» sub-needs



Structural Model — BDD



Brain

- Driver Monitoring ECU — 50 ms cycle, OpenCV / AUTOSAR sw
- Ops: monitorDriver(), classifyState(), selectAlert()

Sensing

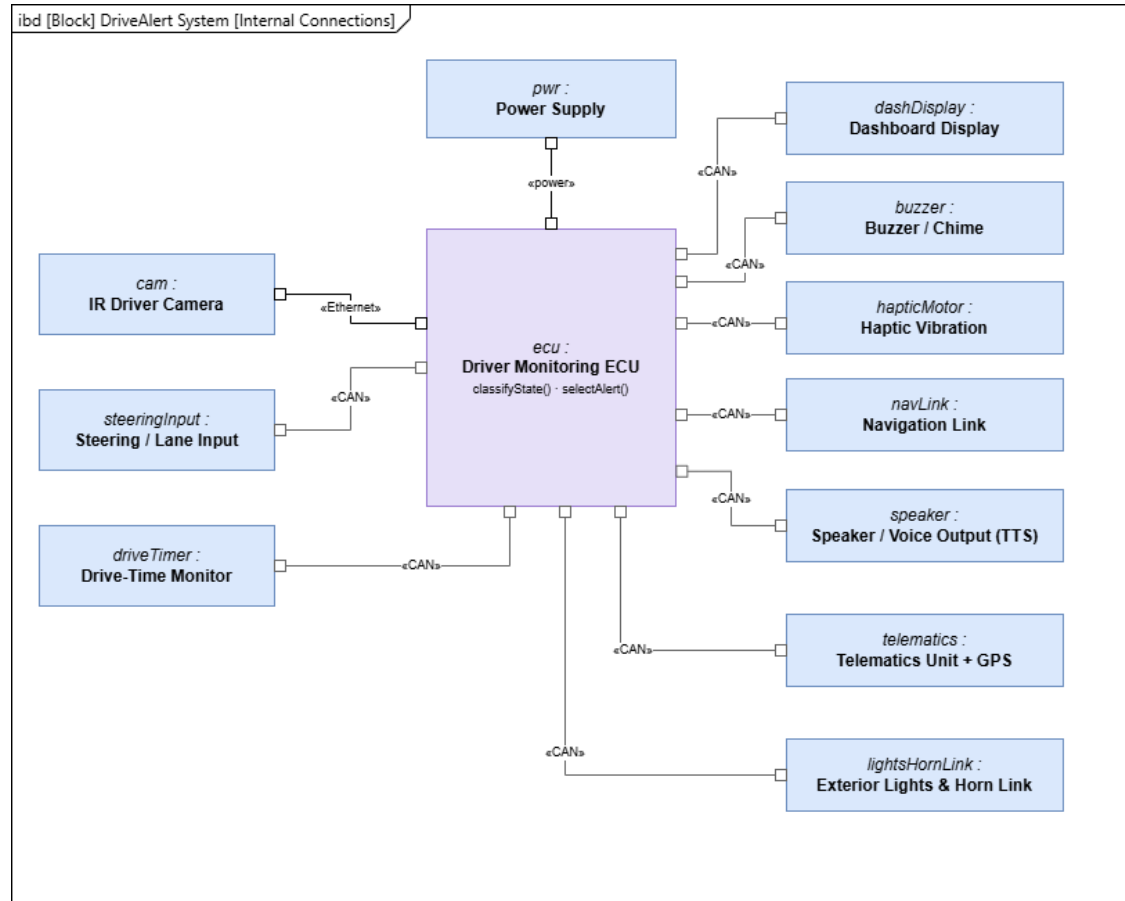
- IR Driver Camera — 1080p · 30 fps · near-IR
- Steering / Lane Input · Drive-Time Monitor

Actuation & comms

- Buzzer · Dashboard · Haptic · Lights & Horn · TTS · Nav · Telematics+GPS
- CAN Bus / Automotive Ethernet · 12 V Power Supply



Structural Model — IBD



IBD — how the blocks are wired through ports and connectors

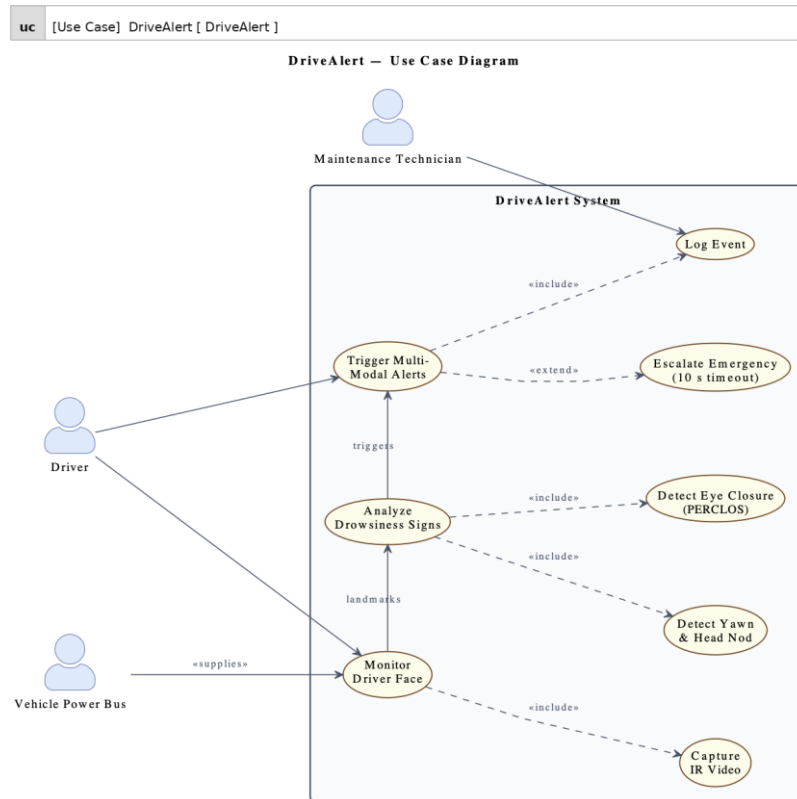
ECU at the centre

- Driver Monitoring ECU hosts `classifyState()` & `selectAlert()`
- All sensing and actuation routes through it

Interfaces & ports

- «Ethernet» — high-bandwidth IR camera feed
- «CAN» — steering, timer, displays, buzzer, haptic, nav, TTS, lights/horn, telematics
- «power» — 12 V supply with watchdog

Use Case Model



Use Case Diagram — actors and the system's externally visible behavior

Actors

- Driver — primary user; receives alerts
- Vehicle Power Bus — «supplies» power
- Maintenance Technician — reads Event Log

Core use cases

- Monitor Driver Face «include» Capture IR Video
- Analyze Drowsiness «include» Detect Eye Closure (PERCLOS) & Detect Yawn / Head-Nod
- Trigger Multi-Modal Alerts «include» Log Event, «extend» Escalate Emergency (10 s timeout)

Behavioral Model — Activity Diagram

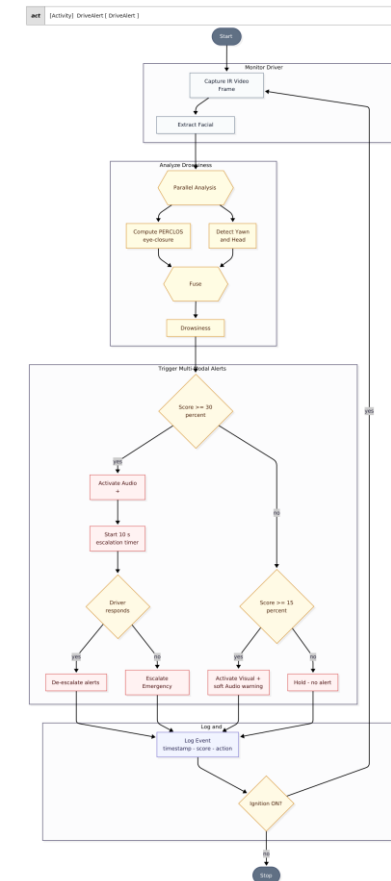
The continuous monitoring loop

Flow (top → bottom)

- Monitor Driver — capture IR frame, extract facial landmarks
- Analyze Drowsiness — parallel: compute PERCLOS + detect yawn / head-nod → fuse → score
- Decision: score $\geq 30\%$?
- Yes → audio + 10 s timer; driver responds? → de-escalate, else escalate emergency
- No → score $\geq 15\%$? visual + soft audio, else hold (no alert)
- Log Event (timestamp · score · action)
- Ignition ON? → loop, else Stop

Why it is shaped this way

- Parallel fork = PERCLOS and yawn/nod evaluated together for a robust score
- Two-level threshold avoids over-warning on mild fatigue
- Every branch logs an event → full audit trail (R-Log)
- Loop is gated by ignition → system sleeps when the car is off



Activity — end-to-end control flow



Behavioral Model — State Machine

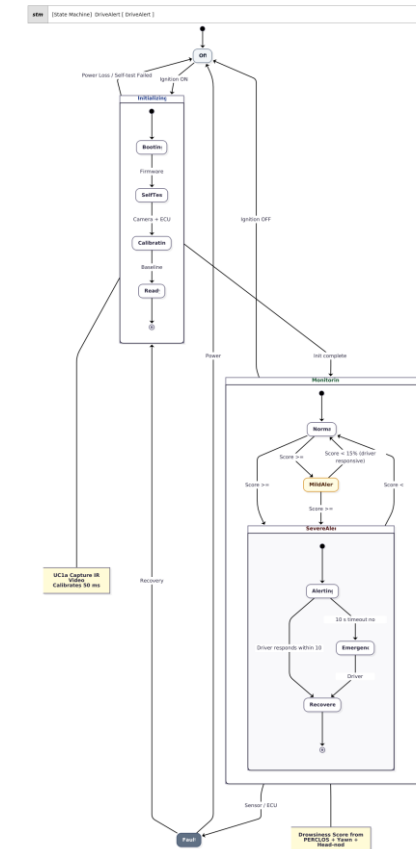
Operating modes of the system

Top-level states

- Off → Ignition ON → Initializing
- Initializing: Booting → SelfTest → Calibrating → Ready
- Monitoring (composite): Normal ⇌ MildAlert ⇌ SevereAlert
- SevereAlert: Alerting → Emergency (10 s timeout) or Recovered (driver responds)
- Fault on sensor/ECU error → Off on recovery

Transition logic

- Score $\geq 15\%$ → MildAlert; score $\geq 30\%$ → SevereAlert
- Score $< 15\%$ (responsive) → back to Normal
- 10 s no-response in SevereAlert → Emergency
- Ignition OFF / power loss / self-test fail → Off
- Nested states keep alert behavior self-contained & safe



State Machine — system modes & transitions



Behavioral Model — Sequence Diagram

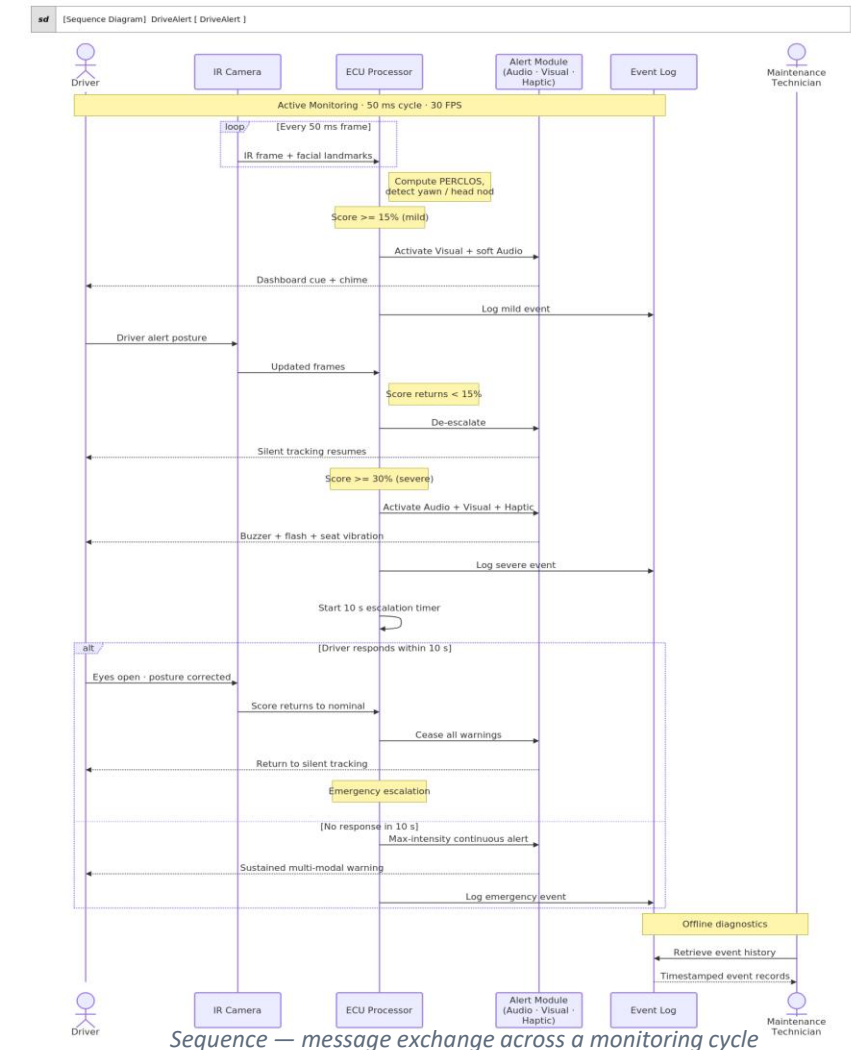
A drowsiness episode, message by message

Lifelines

- Driver
- IR Camera
- ECU Processor
- Alert Module (Audio · Visual · Haptic)
- Event Log
- Maintenance Technician

Scenario walk-through

- loop every 50 ms: frame + landmarks → compute PERCLOS
- Score $\geq 15\%$ (mild) → visual + soft audio → log
- Driver recovers → score $< 15\%$ → de-escalate
- Score $\geq 30\%$ (severe) → audio+visual+haptic → log → start 10 s timer
- alt: responds → cease warnings · no response → max-intensity alert + log emergency
- Offline: technician retrieves event history





Parametric Model — The Governing Math

Constraints that turn signals into decisions

PERCLOS

$$\text{PERCLOS} = (\text{closed_eye_frames} / \text{total_frames}) \times 100$$

Fusion

$$\text{score} = 0.6 \cdot \text{PERCLOS} + 0.25 \cdot \text{yawn_rate} + 0.15 \cdot \text{nod_rate}$$

Thresholds

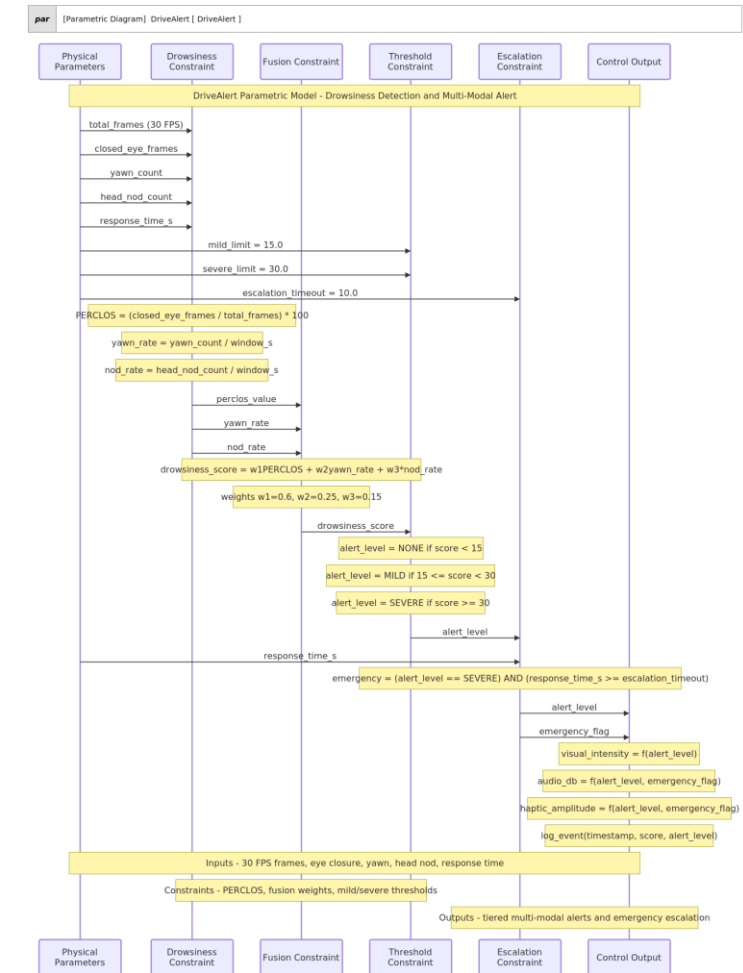
NONE < 15 • MILD 15–30 • SEVERE ≥ 30

Escalation

emergency = (level == SEVERE) AND
(response_time ≥ 10 s)

Outputs

visual_intensity • audio_dB •
haptic_amplitude • log_event(...)



Parametric Diagram — constraints binding inputs to alert outputs

Design Decisions — Justified

Every key choice traces back to a requirement and is reflected consistently across the structural, behavioral and parametric models.



Two-tier thresholds (15% / 30%)

Mild warning corrects most lapses early; severe tier reserved for real danger — cuts false alarms (R1.2, R2).



Multi-modal, escalating alerts

Visual → audio → haptic ensures the cue gets through even if one channel is missed (R2.1–R2.3).



Driver keeps full authority

Advisory only — no steering/brake/engine actuation, so trust and legal liability stay with the human (R4).



10 s emergency escalation

Bounded grace period before eCall + GPS protects an unresponsive driver and other road users (R3).



Weighted sensor fusion

0.6/0.25/0.15 weights favour reliable PERCLOS while yawn & nod add robustness (Parametric).



Fail-safe modes

Self-test, degraded mode and ECU watchdog keep behavior predictable on faults (R5).



Conclusion

- ✓ A complete SysML model of DriveAlert across all four pillars
- ✓ Eight diagrams — requirements, use case, BDD, IBD, activity, sequence, state machine, parametric — form one traceable, consistent model
- ✓ Early, multi-modal warning that keeps the driver in control and escalates only when truly needed
- ✓ Future work: real IR dataset validation, ML-tuned fusion weights, on-vehicle HIL testing

At a glance

4 SysML pillars
8 diagrams
7 requirement groups
13 system blocks
50 ms · 30 FPS loop
15% / 30% / 10 s rules

Thank you

Questions?

